

LECTURE SHEET

EXERCISE-I

Composite, Real valued functions, Even (or) Odd functions,
Inverse of a functions & Inverse of an element

LEVEL-I (MAIN)

Single answer type questions

- If $f(x) = \frac{\cos^2 x + \sin^4 x}{\sin^2 x + \cos^4 x}$ for $x \in R$ then $f(2018) =$
 1) 1 2) 2 3) 3 4) 4
- Suppose that $g(x) = 1 + \sqrt{x}$ and $f(g(x)) = 3 + 2\sqrt{x} + x$, then $f(x)$ is
 1) $1 + 2x^2$ 2) $2 + x^2$ 3) $1 + x$ 4) $2 + x$
- If $g(x) = x^2 + x - 2$ and $\frac{1}{2}gof(x) = 2x^2 - 5x + 2$, then which is $f(x)$
 1) $2x - 3$ 2) $-2x - 3$ 3) $x - 3$ 4) $x + 3$
- If $f(x) = \frac{9^x}{9^x + 3}$ then $f\left(\frac{1}{1996}\right) + f\left(\frac{2}{1996}\right) + \dots + f\left(\frac{1995}{1996}\right) =$
 1) 997 2) 997.5 3) 998 4) 998.5
- If $f(x) = \frac{(x-a)(x-b)}{x}$ and $\frac{f(x)}{(x-y)(x-z)} + \frac{f(y)}{(y-z)(y-x)} + \frac{f(z)}{(z-x)(z-y)} = \frac{k}{xyz}$ then $k =$
 1) a 2) b 3) ab 4) 3ab
- The values of b and c for which the identity $f(x+1) - f(x) = 8x + 3$ is satisfied, where $f(x) = bx^2 + cx + d$, are
 1) $b = 2, c = 1$ 2) $b = 4, c = -1$ 3) $b = -1, c = 4$ 4) $b = -1, c = 1$
- Let $f(x) = \frac{\alpha x}{(x+1)}, x \neq -1$. Then for which value of α is $f(f(x)) = x$?
 1) $\sqrt{2}$ 2) $-\sqrt{2}$ 3) 1 4) -1
- If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin \sqrt{x})^2$, then
 1) $f(x) = \sin^2 x, g(x) = \sqrt{x}$ 2) $f(x) = \sin x, g(x) = |x|$
 3) $f(x) = x^2, g(x) = \sin \sqrt{x}$ 4) f and g cannot be determined
- Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$. Then for all x , $f(g(x))$ is equal to (where $[.]$ represents the greatest integer function)
 1) x 2) 1 3) $f(x)$ 4) $g(x)$

10. If $f: R \rightarrow R$ is defined by $f(x) = x - [x] - \frac{1}{2}$ for $x \in R$, where $[x]$ is the greatest integer not exceeding x , then $\left\{x \in R: f(x) = \frac{1}{2}\right\} =$
- 1) Z 2) N 3) ϕ 4) R
11. If $f: R \rightarrow R$ and $g: R \rightarrow R$ are given by $f(x) = |x|$ and $g(x) = [x]$ for each $x \in R$, then $\{x \in R: g(f(x)) \leq f(g(x))\} =$
- 1) $Z \cup (-\infty, 0)$ 2) $(-\infty, 0)$ 3) Z 4) R
12. If $f(x) = \frac{e^x + e^{-x}}{2}$ then the inverse of $f(x)$ is
- 1) $\log_e(x + \sqrt{x^2 + 1})$ 2) $\log_e \sqrt{x^2 + 1}$
 3) $\log_e(x + \sqrt{x^2 - 1})$ 4) $\log_e(x - \sqrt{x^2 - 1})$
13. If $f(x) = \sin^{-1}\{3 - (x - 6)^4\}^{1/3}$ then $f^{-1}(x) =$
- 1) $6 + \sqrt[3]{3 + \sin^3 x}$ 2) $6 + \sqrt[3]{3 - \sin^3 x}$ 3) $6 + \sqrt[3]{3 + \sin x}$ 4) $6 + \sqrt[3]{3 - \sin x}$
14. If the function $f: R \rightarrow R$ be such that $f(x) = x - [x]$, where $[x]$ denotes the greatest integer less than or equal to x then $f^{-1}(x)$ is
- 1) $\frac{1}{x - [x]}$ 2) $[x] - x$ 3) not defined 4) $x - [x]$
15. Let $f: R \rightarrow R, g: R \rightarrow R$ be two functions given by $f(x) = 2x - 3, g(x) = x^3 + 5$. Then, $(fog)^{-1}$ is equal to
- 1) $\left(\frac{x+7}{2}\right)^{1/3}$ 2) $\left(x - \frac{7}{2}\right)^{1/3}$ 3) $\left(\frac{x-2}{7}\right)^{1/3}$ 4) $\left(\frac{x-7}{2}\right)^{1/3}$
16. If $f: \{1, 2, 3, \dots\} \rightarrow \{0, \pm 1, \pm 2, \dots\}$ is defined by $f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -\left(\frac{n-1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$ then $f^{-1}(-100)$ is
- 1) 100 2) 199 3) 201 4) 200
17. Let $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$, then f is
- 1) an odd function 2) an even function
 3) both odd and even 4) neither odd nor even
18. If $f(x) = ax^5 + bx^3 + cx + d$ is odd then
- 1) $a = 0$ 2) $b = 0$ 3) $c = 0$ 4) $d = 0$

19. If the graph of the function $f(x) = \frac{a^x - 1}{x^a(a^2 + 1)}$ is symmetrical about the y-axis, then 'n' equals
- 1) 2 2) $\frac{2}{3}$ 3) $\frac{1}{4}$ 4) 3

LEVEL-II (ADVANCED)

Single answer type questions

- Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is

a) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ b) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$

c) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$ d) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
- If $f\left(2x + \frac{y}{8}, 2x - \frac{y}{8}\right) = xy$, then $f(m, n) + f(n, m) = 0$

a) only when $m = n$ b) only when $m \neq n$

c) only when $m = -n$ d) for all m and n
- For $x \in \mathbb{R}, x \neq 0, x \neq 1$, let $f_0(x) = \frac{1}{1-x}$ and $f_{n+1}(x) = f_0(f_n(x)), n = 0, 1, 2, \dots$. Then the value of $f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to :

a) $\frac{8}{3}$ b) $\frac{5}{3}$ c) $\frac{4}{3}$ d) $\frac{1}{3}$
- If $f(x) = \begin{cases} |x|, & x \leq 1 \\ 2-x, & x > 1 \end{cases}$ then $f(f(x)) =$

a) $\begin{cases} 2-|x|, & x < -1 \\ |x|, & -1 \leq x \leq 1 \\ |2-x|, & x > 1 \end{cases}$ b) $\begin{cases} |x|, & x < -1 \\ 2-|x|, & -1 \leq x \leq 1 \\ |2-x|, & x > 1 \end{cases}$ c) $\begin{cases} |2-x|, & x < -1 \\ |x|, & -1 \leq x \leq 1 \\ 2-|x|, & x > 1 \end{cases}$ d) Cannot say
- Let $f(x) = ([a]^2 - 5[a] + 4)x^3 - (6\{a\}^2 - 5\{a\} + 1)x - (\tan x) \times \operatorname{sgn} x$ be an even function for all $x \in \mathbb{R}$. Then the sum of all possible values of a is (where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function, respectively)

a) $\frac{17}{6}$ b) $\frac{53}{6}$ c) $\frac{31}{3}$ d) $\frac{35}{3}$
- A function whose graph is symmetrical about the y-axis is given by

a) $f(x) = \sin\left[\log\left(x + \sqrt{x^2 + 1}\right)\right]$ b) $f(x) = \frac{\sec^4 x + \operatorname{cosec}^4 x}{x^3 + x^4 \cot x}$

c) $f(x+y) = f(x) + f(y) \forall x, y \in \mathbb{R}$ d) $f(x) = x^2$

16. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g(f(x))$ is invertible in the domain

- a) $\left[0, \frac{\pi}{2}\right]$ b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ d) $[0, \pi]$

17. Let $f(x) = (x+1)^2 - 1$, $x \geq -1$. Then the set $\{x : f(x) = f^{-1}(x)\}$ is

- a) $\left\{0, -1, \frac{-3+i\sqrt{3}}{2}, \frac{-3-i\sqrt{3}}{2}\right\}$ b) $\{0, 1, -1\}$
c) $\{0, -1\}$ d) empty

18. Let $f(x) = (x+1)^2 - 1$, $x \geq -1$

Statement 1 : The set $\{x : f(x) = f^{-1}(x)\} = \{0, -1\}$.

Statement 2 : f is a bijection.

- a) Statement 1 is true, statement 2 is true; statement 2 is a correct explanation for statement 1.
b) Statement 1 is true, statement 2 is true; statement 2 is not a correct explanation for statement 1.
c) Statement 1 is true, statement 2 is false
d) Statement 1 is false, statement 2 is true

19. Let ' f ' be an injective mapping with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$ such that exactly one of the following statements is correct and the remaining are false $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$ then $f^{-1}(1) =$

- a) x b) y c) z d) 2

More than one correct answer type questions

20. Let $f(x) = \ln(2x - x^2) + \sin\left(\frac{\pi x}{2}\right)$ then

- a) graph of ' f ' is symmetrical about the line $x = 1$ b) graph of ' f ' is symmetrical about the line $x = 2$
c) max. value of ' f ' is 1 d) min. value of ' f ' does not exist

21. Let $f: R - \{1\} \rightarrow R - \{2\}$ defined by $f(x) = \frac{2x+3}{x+1}$ be a bijection mark all the correct alternative w.r.t real solutions

- a) the number of real solutions of $f(x) = f^{-1}(x)$ is 2
b) $f(x) = f^{-1}(x)$, if $f(x) = x$
c) $f(x) = f^{-1}(x)$, only if $f(x) = x$ d) All the above are correct

22. Let f be a function defined by $f(x) = \frac{x-5}{x-3}$, $x \neq 3, 2$ $f^k(x)$ denote the composition of f with itself taken k time i.e. $f^3(x) = f(f(f(x)))$ then

- a) $f^{2012}(2009) = 2009$ b) $f^{2009}(2010) = \frac{2005}{2007}$
c) $f^{2009}(2009) = \frac{1002}{1003}$ d) $f^{2012}(2012) = 2012$

23. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2 \theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then the value of

$f\left(\frac{1}{3}\right)$ is (are)

a) $1 - \sqrt{\frac{3}{2}}$

b) $1 + \sqrt{\frac{3}{2}}$

c) $1 - \sqrt{\frac{2}{3}}$

d) $1 + \sqrt{\frac{2}{3}}$

24. If $f(x) = \frac{(x-3)(x+2)(x+5)}{(x+1)(x-7)}$

a) $f(x) > 0 \forall x \in (-5, -2) \cup (-1, 3) \cup (7, \infty)$

b) $f(x) < 0 \forall x \in (-\infty, -5) \cup (-2, -1) \cup (3, 7)$

c) $f(x) > 0 \forall x \in (-5, -2) \cup (-1, 3) \cup (9, \infty)$

d) $f(x) < 0 \forall x \in (-\infty, -5) \cup (-2, -1) \cup (3, 4)$

25. Which of the following pairs of functions is/are identical ?

a) $f(x) = \tan(\tan^{-1} x)$ and $g(x) = \cot(\cot^{-1} x)$

b) $f(x) = \operatorname{sgn}(x)$ and $g(x) = \operatorname{sgn}(\operatorname{sgn}(x))$

c) $f(x) = \cot^2 x \cdot \cos^2 x$ and $g(x) = \cot^2 x - \cos^2 x$

d) $f(x) = e^{\ln \sec^{-1} x}$ and $g(x) = \sec^{-1} x$

26. $f(x) = \tan \frac{x}{2} \sec x + \tan \frac{x}{2^2} \sec \frac{x}{2} + \tan \frac{x}{2^3} \sec \frac{x}{2^2} + \dots + \tan \frac{x}{2^n} \sec \frac{x}{2^{n-1}}$ and $g(x) = f(x) + \tan\left(\frac{x}{2^n}\right)$ where $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $n \in \mathbb{N}$ then

a) $g(x)$ is an even function

b) $g(x)$ is an odd function

c) $g(x)$ is neither even nor odd

d) $g(x)$ is both even and odd

27. If $y = f(x) = \frac{(x+2)}{(x-1)}$, then

a) $x = f(y)$

b) $f(1) = 3$

c) y increases with x for $x < 1$

d) f is a rational function of x

Linked comprehension type questions

Passage - I :

Let $f(x) = \log_e(x + \sqrt{x^2 + 1})$, domain of f is where $f(x)$ is defined for real values of x . If f is bijective then $f^{-1}(x)$ exists

28. $f^{-1}(x)$ is defined on

a) $(0, \infty)$

b) $(-\infty, \infty)$

c) $[0, e]$

d) $(-\infty, 0)$

29. The inverse of f is positive on

a) $(0, \infty)$

b) $(-\infty, \infty)$

c) $[0, e]$

d) $(-\infty, 0)$

Matrix matching type questions

30. If $f: R \rightarrow R$ is defined by

$$f(x) = \begin{cases} x+4 & \text{for } x < -4 \\ 3x+2 & \text{for } -4 \leq x < 4 \\ x-4 & \text{for } x \geq 4 \end{cases} \text{ then the correct matching of List I from List II is}$$

List - I

- A) $f(-5) + f(-4)$
 B) $f(|f(-8)|)$
 C) $f(f(-7) + f(3))$
 D) $f(f(f(f(0)))) + 1$

List - II

- i) 14
 ii) 4
 iii) -11
 iv) -1
 v) 1
 vi) 0

31. Match the following

Column-I

- A) If $f(x) = e^{2x}$ and $g(x) = \log(x)$ then $(f \circ g)(x)$
 B) If $f(x) = \frac{x}{x+1}$, $g(x) = \frac{x}{1-x}$ then $x \circ (f \circ g)(x)$
 C) If $f(x) = \frac{ax+b}{cx+d}$ then $(f \circ g)(x) = x$ then $a+d$
 D) If $g(x) = 1 + \sqrt{x}$ and $f\{g(x)\} = 3 + 2\sqrt{x} + x$ then $f(x) =$

Column-II

- p) 0
 q) $x^2 + 2$
 r) x^2
 s) 1

32. Match the following :

Column-I

- A) If $f(x) = x+1$, when $-1 \leq x < 0$
 $= x^2 - 1$; when $x \geq 0$ then $(f \circ f)(x)$ for $-1 \leq x < 0$ is
 B) If $f\left(\frac{2 \tan x}{1 + \tan^2 x}\right) = \frac{(\cos(2x) + 1)(\sec^2 x + 2 \tan x)}{2}$ then $f(x)$ is
 C) If $f(x+y+1) = (\sqrt{f(x)} + \sqrt{f(y)})^2 \forall x,$
 $y \in R$ and $f(0) = 1$ then $f(x)$ is
 D) If $4 < x < 5$ and $f(x) = \left[\frac{x}{4}\right] + 2x + 2$
 (Where $[.]$ is G.I.F) then $f^{-1}(x)$ is

Column-II

- p) $\frac{x-3}{2}$
 q) $x^2 + 2x$
 r) $1 + x$
 s) $(x+1)^2$

33. Let $f(x) = \log \frac{1+x}{1-x}$ and $g(x) = \frac{3x+x^3}{1+3x^2}$, then

Column-I

- A) $\log \frac{1-e}{1+e}$
 B) $\log \left(\frac{e-1}{1+e} \right)$
 C) $\log(0)$
 D) $\log \left(\frac{e-1}{e+1} \right)$

Column-II

- p) 3
 q) -3
 r) 0
 s) 1

Integer answer type questions

34. If $f^3(x) - 3f^2(x) + 3f(x) - 1 = x^6$ then the value of $f(0)$ is

EXERCISE-II

Domain and Range

LEVEL-I (MAIN)

Single answer type questions

- For which Domain, the functions $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ are equal to
 1) $\mathbb{R}$ 2) $\left\{ \frac{1}{2}, -2 \right\}$ 3) $\left(\frac{1}{2}, 2 \right)$ 4) $\left[\frac{1}{2}, 2 \right]$
- The domain of the function $f(x) = \frac{1}{\sqrt{4x - |x^2 - 10x + 9|}}$ is
 1) $(7 - \sqrt{40}, 7 + \sqrt{40})$ 2) $(0, 7 + \sqrt{40})$
 3) $(7 - \sqrt{40}, \infty)$ 4) $(7 - \sqrt{40}, 3) \cup (3, 7 + \sqrt{40})$
- The domain of definition of the function $f(x)$ given by the equation $2^x + 2^y = 2$ is
 1) $0 < x \leq 1$ 2) $0 \leq x \leq 1$ 3) $-\infty < x \leq 0$ 4) $-\infty < x < 1$
- The domain of the function $f(x) = \sqrt{1 - \sqrt{1 - \sqrt{1 - x^2}}}$ is
 1) $\{x | x < 1\}$ 2) $\{x | x > -1\}$ 3) $[0, 1]$ 4) $[-1, 1]$
- The domain of the function $f(x) = \left[\log_{10} \left(\frac{5x - x^2}{4} \right) \right]^{1/2}$ is
 1) $-\infty < x < \infty$ 2) $1 \leq x \leq 4$ 3) $4 \leq x \leq 16$ 4) $-1 \leq x \leq 1$
- The domain of $f(x) = \sqrt{\frac{\log_{0.3} |x-2|}{|x|}}$ is
 1) $[1, 2]$ 2) $[2, 3]$ 3) $[1, 2) \cup (2, 3]$ 4) $[0, 3]$

7. The domain of $f(x) = \cos^{-1}\left(\frac{2}{2+\sin x}\right)$ contained in $[0, 2\pi]$ is
- 1) $\left[0, \frac{\pi}{2}\right]$ 2) $\left[\frac{\pi}{2}, \pi\right]$ 3) $[0, \pi]$ 4) $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$
8. The largest interval lying in $(-\pi/2, \pi/2)$ for which $f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$ is defined is
- 1) $[0, \pi]$ 2) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ 3) $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ 4) $\left[0, \frac{\pi}{2}\right)$
9. The function $f(x) = \cot^{-1}\left(\sqrt{(x+3)x}\right) + \cos^{-1}\left(\sqrt{x^2+3x+1}\right)$ is defined on the set S , then S is equal to
- 1) $\{-3, 0\}$ 2) $[-3, 0]$ 3) $[0, 3]$ 4) $(-3, 0)$
10. The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is
- 1) $[2, 3]$ 2) $[2, 3)$ 3) $[1, 2]$ 4) $[1, 2)$
11. The domain of the function $f(x) = \frac{\sin^{-1}(3-x)}{\ln(|x|-2)}$ is
- 1) $[2, 4]$ 2) $(2, 3) \cup (3, 4]$ 3) $[2, \infty)$ 4) $(-\infty, -3) \cup [2, \infty)$
12. The domain of $f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right) + [\log(3-x)]^{-1}$ is
- 1) $[-2, 6]$ 2) $[-6, 2) \cup (2, 3)$ 3) $[-6, 2]$ 4) $[-2, 2] \cup (2, 3)$
13. The range of the function $f(x) = \frac{2+x}{2-x}, x \neq 2$ is
- 1) $\mathbb{R}$ 2) $\mathbb{R} - \{-1\}$ 3) $\mathbb{R} - \{1\}$ 4) $\mathbb{R} - \{2\}$
14. The range of $f(x) = \frac{1-\tan x}{1+\tan x}$ is
- 1) $(-\infty, \infty)$ 2) $(-\infty, 0)$ 3) $(0, \infty)$ 4) $(-\infty, -1) \cup (-1, \infty)$
15. The range of the function $f(x) = {}^{7-x}P_{x-3}$ is
- 1) $\{1, 2, 3\}$ 2) $\{1, 2, 3, 4, 5\}$ 3) $\{1, 2, 3, 4\}$ 4) $\{1, 2, 3, 4, 5, 6\}$
16. The range of the function $f(x) = \frac{e^x - e^{|x|}}{e^x + e^{|x|}}$ is
- 1) $(-\infty, \infty)$ 2) $[0, 1]$ 3) $(-1, 0]$ 4) $(-1, 1)$

17. Range of the function $\frac{x^3 - 27}{x - 3}$
- 1) $\left[\frac{27}{4}, \infty\right)$ 2) $\left(-\infty, \frac{27}{4}\right]$ 3) $\left[\frac{4}{27}, \infty\right)$ 4) $\left(-\infty, \frac{4}{27}\right]$
18. The range of $f(x) = \frac{x^4}{1+x^8}$ is
- 1) $[0, \infty)$ 2) $\left[0, \frac{1}{2}\right]$ 3) $[0, 1]$ 4) $(-\infty, \infty)$
19. The range of $f(x) = [\sin x] + [\cos x]$, where $[.]$ denotes the greatest integer function, is
- 1) $\{0\}$ 2) $\{0, 1\}$ 3) $\{1\}$ 4) $\{1, 2\}$
20. If $f(x) = ax^7 + bx^3 + cx - 5$, a, b, c are real constants, and $f(-7) = 7$, then the range of $f(7) + 17 \cos x$ is
- 1) $[-34, 0]$ 2) $[0, 34]$ 3) $[-34, 34]$ 4) $[34, \infty)$
21. The range of the function $f(x) = |x-1| + |x-2|$, $-1 \leq x \leq 3$, is
- 1) $[1, 3]$ 2) $[1, 5]$ 3) $[3, 5]$ 4) $[2, 5]$
22. Range of the function $f(x) = \sqrt{[x]} - x$ is
- 1) $\mathbb{R}$ 2) $\{1\}$ 3) $\{0\}$ 4) $(0, \infty)$
23. If the function $f: \mathbb{R} \rightarrow B$ defined by $f(x) = \sqrt{x^2}$ is surjective then B is
- 1) $(-\infty, 0]$ 2) $[0, \infty)$ 3) $(0, \infty)$ 4) $\mathbb{R}$

LEVEL-II (ADVANCED)

Single answer type questions

1. The domain of the function $f(x) = \sin^{-1} \left\{ \log_2 \frac{x^2}{2} \right\}$ is given by _____
- a) $[-2, -1] \cup [1, 2]$ b) $[-2, 0] \cup [0, 2]$
 c) $[-3, -1] \cup [1, 2]$ d) $[-2, 2]$
2. The domain of definition of the function $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ for real-valued x is
- a) $\left[-\frac{1}{4}, \frac{1}{2}\right]$ b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ c) $\left(-\frac{1}{2}, \frac{1}{9}\right)$ d) $\left[-\frac{1}{4}, \frac{1}{4}\right]$
3. The function $f(x) = \sqrt{\ln_{(b+1)}(x^2 + 4x + 4)}$ is

4. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g(f(x))$ is invertible in the domain
- a) $\left[0, \frac{\pi}{2}\right]$ b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ d) $[0, \pi]$
5. Let f be a function with domain $[-3, 5]$ and let $g(x) = |3x + 4|$. Then, the domain of $(fog)(x)$ is
- a) $\left(-3, \frac{1}{3}\right)$ b) $\left[-3, \frac{1}{3}\right]$ c) $\left[-3, \frac{1}{3}\right]$ d) $\left[-3, -\frac{1}{3}\right]$
6. The domain of $f(x) = \sin^{-1}[2 - 4x^2]$ (where $[.]$ is G.I.F) is
- a) $\left[\frac{-\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]$ b) $\left[\frac{-3}{2}, \frac{3}{2}\right] - \{0\}$ c) $\left[\frac{-\sqrt{3}}{2}, 0\right] \cup \left(0, \frac{\sqrt{3}}{2}\right]$ d) ϕ
7. Domain of $\frac{1}{\sqrt{[x]^2 - [x] - 2}}$
- a) $R - [-1, 3]$ b) $(-\infty, -3) \cap [3, \infty)$ c) $[2, \infty)$ d) $(-\infty, 3]$
8. If $f(x) = \log_{[x-1]} \left(\frac{[x]}{x}\right)$, where $[.]$ is G.I.F then domain and range are
- a) $(2, \infty), (0, 1)$ b) $[3, \infty), \{0\}$ c) $[3, \infty), \{0, 1\}$ d) $(-\infty, \infty); \{0\}$
9. If $f(x) = \cos^{-1}\left(x - x^2\right) + \sqrt{1 - \frac{1}{[x]}} + \frac{1}{[x^2 - 1]}$ then domain of $f(x)$ (where $[.]$ G.I.F) is
- a) $\left[\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$ b) $\left(-\sqrt{2}, \frac{1-\sqrt{5}}{2}\right)$ c) $\left[\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$ d) $\left[\sqrt{7}, \frac{\sqrt{7}}{2}\right]$
10. The domain of $f(x) = \sqrt{2\{x\}^2 - 3\{x\} + 1}$, where $\{.\}$ denotes the fractional part in $[-1, 1]$ is
- a) $[-1, 1] \sim \left(\frac{1}{2}, 1\right)$ b) $\left[-1, -\frac{1}{2}\right] \cup \left[0, \frac{1}{2}\right] \cup \{1\}$
- c) $\left[-1, \frac{1}{2}\right]$ d) $\left[-\frac{1}{2}, 1\right]$
11. The domain of the function $f(x) = \frac{1}{\sqrt{\{\sin x\} + \{\sin(\pi + x)\}}}$, (where $\{.\}$ denotes the fractional part) is
- a) $[0, \pi]$ b) $(2n+1)\pi/2, n \in Z$ c) $(0, \pi)$ d) $\left(0, \frac{\pi}{2}\right)$
12. The domain of definition of the function $f(x) = \{x\}^{\{x\}} + [x]^{\{x\}}$ is $\{.\}$ represents fractional part and $[.]$ represents greatest integral part
- a) $R - 1$ b) $R - [0, 1]$ c) $R - \{I \cup (0, 1)\}$ d) $I \cup (0, 1)$
13. Let $h(x) = [kx + 5]$, the domain of $f(x)$ be $[-5, 7]$, the domain of $f(h(x))$ be $[-6, 1]$ and the range of $h(x)$ be the same as the domain of $f(x)$. Then the value of k is
- a) 1 b) 2 c) 3 d) 4

14. The values of $f(x) = 3\sin\left(\sqrt{\frac{\pi^2}{16} - x^2}\right)$ lie in the interval _____
- a) $\left[0, \frac{3}{\sqrt{2}}\right]$ b) $\left[-\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right]$ c) $\left(0, \frac{3}{\sqrt{2}}\right)$ d) $\left(-\frac{3}{\sqrt{2}}, 0\right)$
15. If $f(x) = \sin\left(\log_e\left\{\frac{\sqrt{4-x^2}}{1-x}\right\}\right)$, then the domain of $f(x)$ is _____ and its range is _____
- a) R and $[0, 1]$ b) $(-2, 1)$ and $[-1, 1]$ c) $(0, \infty)$ and $[-1, 1]$ d) $(0, \infty)$ and $[0, 1]$
16. If $x \in R$ and $P = \frac{x^2}{x^4 - 2x^2 + 4}$, then P lies interval
- a) $\left[0, \frac{1}{2}\right]$ b) $\left[\frac{3}{4}, \frac{4}{5}\right]$ c) $\left[0, \frac{1}{3}\right]$ d) $\left[0, \frac{1}{4}\right]$
17. The range of $f(x) = \sqrt{(1-\cos x)}\sqrt{(1-\cos x)}\sqrt{(1-\cos x)}\sqrt{\dots\infty}$ is
- a) $[0, 1]$ b) $[0, 1/2]$ c) $[0, 2]$ d) $\left[\frac{1}{2}, 1\right]$
18. Let $f(x) = \sqrt{|x| - \{x\}}$ (where $\{.\}$ denote the fractional part of x and x, y are its domain and range, respectively). Then
- a) $x \in \left(-\infty, \frac{1}{2}\right]$ and $y \in \left[\frac{1}{2}, \infty\right)$ b) $x \in \left(-\infty, -\frac{1}{2}\right] \cup [0, \infty)$ and $y \in \left[\frac{1}{2}, \infty\right)$
- c) $x \in \left(-\infty, -\frac{1}{2}\right] \cup [0, \infty)$ and $y \in [0, \infty)$ d) $x \in \left(-\infty, \frac{1}{2}\right)$ and $y \in \left[\frac{-1}{2}, \infty\right)$
19. The range of $f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right)$ is
- a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ b) $[0, \pi]$ c) $\left[0, \frac{\pi}{2}\right]$ d) $(0, \pi)$
20. Let $f(x) = \sin x$ and $g(x) = \log_e |x|$. If the ranges of the composition functions $f \circ g$ and $g \circ f$ are R_1 and R_2 , respectively, then
- a) $R_1 = \{u: -1 \leq u < 1\}, R_2 = \{v: -\infty < v < 0\}$ b) $R_1 = \{u: -\infty < u < 0\}, R_2 = \{v: -\infty < v < 0\}$
- c) $R_1 = \{u: -1 < u < 1\}, R_2 = \{v: -\infty < v < 0\}$ d) $R_1 = \{u: -1 \leq u \leq 1\}, R_2 = \{v: -\infty < v \leq 0\}$
21. The range of $f(x) = \sin^{-1}\left[\frac{1}{2} + x^2\right]$ is [.] denotes greatest integer function)
- a) $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$ b) $\left\{0, \frac{\pi}{2}\right\}$ c) $\left\{\frac{\pi}{2}\right\}$ d) $\{0, \pi\}$

22. Let $f(x) = a + b \cos^{-1}x (b > 0)$. If domain and range of $f(x)$ are the same set then $b - a$ is equal to

- a) $1 - \frac{2}{\pi}$ b) $1 + \frac{2}{\pi}$ c) 2 d) $1 - \frac{1}{\pi}$

23. The range of $f(x) = [1 + \sin x] + [2 + \sin \frac{x}{2}] + [3 + \sin \frac{x}{3}] + \dots + [n + \sin \frac{x}{n}] \forall x \in [0, \pi]$, where $[.]$ where denotes the greatest integer function, is

- a) $\left\{ \frac{n^2 + n - 2}{2}, \frac{n(n+1)}{2} \right\}$ b) $\left\{ \frac{n(n+1)}{2} \right\}$
c) $\left\{ \frac{n^2 + n - 2}{2}, \frac{n(n+1)}{2}, \frac{n^2 + n + 2}{2} \right\}$ d) $\left\{ \frac{n(n+1)}{2}, \frac{n^2 + n + 2}{2} \right\}$

More than one correct answer type questions

24. For the function $f(x)$ satisfying $2, f(\sin x) + f(\cos x) = x, \forall x \in R$

- a) Domain is $[0, 1]$ b) range is $\left[\frac{-2\pi}{3}, \frac{\pi}{3} \right]$
c) Domain is $[-1, 1]$ d) range is $\left[\frac{-\pi}{2}, \frac{\pi}{2} \right]$

25. If $e^x + e^{f(x)} = e$, then for $f(x)$

- a) domain = $(-\infty, 1)$ b) range = $(-\infty, 1)$
c) domain = $(-\infty, 0]$ d) range = $(-\infty, 1]$

26. Let $f(x) = \left[\left[\sqrt{n^2 + 1} \right] - \left[\sqrt{n^2 + x} \right] \right]^{1/2}$, where $[.]$ is the greatest inger function and $n \in N$ then the value of x for which $f(x)$ is defined

- a) $\sqrt{k}$, where $k = 0, 1, 2, \dots, n$ b) $[-n^2, 2n+1)$
c) $[0, 2n+1)$ d) N

27. Domain of $f(x) = \sin^{-1} [2 - 4x^2]$ is ($[.]$ denotes the greatest integer function)

- a) $\left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2} \right] - \{0\}$ b) $\left[-\frac{\sqrt{3}}{2}, 1 \right)$ c) $\left[-\frac{\sqrt{3}}{2}, 0 \right) \cup \left(0, \frac{\sqrt{3}}{2} \right]$ d) $\left[-\frac{\sqrt{3}}{2}, 8 \right]$

28. Let $f(x) = \sec^{-1} [1 + \cos^2 x]$, where $[.]$ denotes the greatest integer function. Then the

- a) domain of f is R b) domain of f is $[1, 2]$
c) range of f is $[2, 1]$ d) range of f is $\{\sec^{-1}1, \sec^{-1}2\}$

29. If $f(x) = \sqrt{3|x| - x - 2}$ and $g(x) = \sin x$ then the domain of $(f \circ g)(x) =$

- a) $\left\{2m\pi + \frac{\pi}{2}\right\}; m \in \mathbb{Z}$ b) $\left[2m\pi + \frac{7\pi}{6}, 2m\pi + \frac{11\pi}{6}\right]; m \in \mathbb{Z}$
 c) $\left\{2m\pi + \frac{\pi}{3}\right\}; m \in \mathbb{Z}$ d) ϕ

Linked comprehension type questions

Passage - I :

$$f(x) = \begin{cases} x+1 & x \leq 1 \\ 2x+1 & 1 < x \leq 2 \end{cases} \text{ and } g(x) = \begin{cases} x^2, & -1 \leq x < 2 \\ x+2, & 2 \leq x \leq 3 \end{cases}$$

30. The domain of the function $f(g(x))$ is

- a) $[0, \sqrt{2}]$ b) $[-1, 2]$ c) $[-1, \sqrt{2}]$ d) $[\sqrt{2}, 2]$

31. The range of the function $f(g(x))$ is

- a) $[1, 5]$ b) $[2, 3]$ c) $[1, 2] \cup (3, 5]$ d) $[1, 4]$

32. The number of roots of the equation $f(g(x)) = 2$ is

- a) 1 b) 2 c) 4 d) 3

Passage - II :

Let $p(x)$ be a quadratic polynomial with real coefficient such that for all real 'x' the relation $2(1 + p(x)) = p(x-1) + p(x+1)$ holds of $p(0) = 8$ and $p(2) = 32$ then

33. Sum of all the coefficients of $p(x)$ is

- a) 15 b) 17 c) 19 d) 21

34. If the range of $p(x)$ in $[m, \infty)$ then the value of 'm' is

- a) -5 b) -12 c) -15 d) -17

Passage - III :

A rational function is defined as quotient of two polynomials $p(x)$ and $q(x)$. The domain of the rational function must be all reals except the roots of the equation $q(x) = 0$. The range of rational functions can be found by finding minimum and maximum values of the function. In case $p(x)$ and $q(x)$ have a common factor $x - \beta$. Then after cancelling the common factor. The rational function must assume a value of $x = \beta$ which should be delete from the range since β is not in the domain of the rational function

35. The range of the rational function $f(x) = \frac{3x+1}{2x+1}$ must be

- a) $R - \left\{-\frac{1}{2}\right\}$ b) $R - \left\{-\frac{1}{3}\right\}$ c) $R - \left\{+\frac{3}{2}\right\}$ d) R

36. The range of the rational function $f(x) = \frac{2x+1}{2x^2+5x+2}$ must be

- a) $R - \{0\}$ b) $R - \{-2\}$ c) $R - \{0, -2\}$ d) $R - \{0, 2/3\}$

37. The range of rational function $f(x) = \frac{2x^2 + 5x + 2}{2x + 1}$

- a) $R - \{0\}$ b) $R - \{-2\}$ c) $R - \left\{0, -2, \frac{2}{3}\right\}$ d) $R - \left\{\frac{3}{2}\right\}$

Passage - IV :

Let $f(x) = \begin{cases} 2x + a, & x \geq -1 \\ bx^2 + 3, & x < -1 \end{cases}$ and $g(x) = \begin{cases} x + 4, & 0 \leq x \leq 4 \\ -3x - 2, & -2 < x < 0 \end{cases}$

38. $g(f(x))$ is not defined if

- a) $a \in (10, \infty), b \in (5, \infty)$ b) $a \in (4, 10), b \in (5, \infty)$
c) $a \in (10, \infty), b \in (0, 1)$ d) $a \in (4, 10), b \in (1, 5)$

39. If the domain of $g(f(x))$ is $[-1, 4]$, then

- a) $a = 1, b > 5$ b) $a = 2, b > 7$ c) $a = 2, b > 10$ d) $a = 0, b \in R$

40. If $a = 2$ and $b = 3$, then the range of $g(f(x))$ is

- a) $(-2, 8]$ b) $(0, 8]$ c) $[4, 8]$ d) $[-1, 8]$

Integer answer type questions

41. If the domain of $f(x)$ is $[0, 5]$, then the number of integers in the domain of $f(\log_2(x^2 + 2x - 3))$ is _____

42. If the range of the function $f(x) = \frac{(1+x+x^2)(1+x^4)}{x^3}$ when $x > 0$ is $[K, \infty)$ then K is

EXERCISE-III

Nature of functions, Functional equations, Piecise functions, No. of functions

LEVEL-I (MAIN)

Single answer type questions

- Let R be the set of real numbers. If $f: R \rightarrow R$ is a function defined by $f(x) = x^2$, then f is
 - injective but not surjective
 - surjective but not injective
 - bijjective
 - neither injective nor surjective
- Let the function $f: R \rightarrow R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is
 - one-to-one and onto
 - one-to-one but not onto
 - onto but not one-to-one
 - neither one-to-one nor onto
- Let $f: R \rightarrow R$ is defined by $f(x) = 2x^3 + 2x^2 + 300x + 5\sin x$, then f is
 - one-one onto
 - one-one into
 - many one onto
 - many one into
- Function $f: (-\infty, -1] \rightarrow (0, e^5]$ defined by $f(x) = e^{x^3 - 3x + 2}$ defined by is
 - many-one and onto
 - many-one and into
 - one-one and onto
 - one-one and into

5. If $f(x) = \frac{\sin([x]\pi)}{x^2 + x + 1}$, where $[.]$ denotes the greatest integer function, then
- f is one-one
 - f is one-one and non-constant
 - f is a constant function
 - f is onto
6. Let $n(A) = 4$ and $n(B) = k$. The number of all possible injections from A to B is 120 then $k =$
- 9
 - 24
 - 5
 - 6
7. Let $A = \{1, 2, 3, \dots, n\}$ and $B = \{1, b, c\}$, the number of functions from A to B that are onto is
- $3^n - 2^n$
 - $3^n - 2^n - 1$
 - $3(2^n - 1)$
 - $3^n - 3(2^n - 1)$
8. The number of non-surjective mappings that can be defined from $A = \{1, 4, 9, 1, 6\}$ to $B = \{2, 8, 16, 32, 64\}$ is
- 1024
 - 20
 - 505
 - 625
9. The number of one-one onto functions that can be defined from $\{1, 2, 3, 4\}$ onto set B is 24 then $n(B) =$
- 4
 - 2
 - 3
 - 6
10. $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d, e\}$, then the number of all possible constant functions from A to B is
- 9
 - 4
 - 5
 - 16
11. The number of linear functions which map $[-1, 1]$ to $[0, 2]$ are
- One
 - Two
 - Four
 - Three
12. If $f(x)$ is an odd periodic function with period 2, then $f(4) =$
- 4
 - 4
 - 2
 - 0
13. The period of the function $f(x) = c^{\sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cos\left(x + \frac{\pi}{3}\right)}$ is (where c is constant)
- 1
 - $\frac{\pi}{2}$
 - π
 - Cannot be determined
14. The period of $f(x) = e^{x - [x] + \cos(\pi x) + \cos(2\pi x) + \dots + \cos(n\pi x)}$ (where $[.]$ is G.I.F) is
- 1
 - $\frac{1}{2}$
 - $\frac{1}{4}$
 - $\frac{1}{5}$
15. If $f\left(x + \frac{1}{2}\right) + f\left(x - \frac{1}{2}\right) = f(x)$ for all $x \in \mathbb{R}$, then the period of $f(x)$ is
- 1
 - 2
 - 3
 - 4
16. If $f(3x + 2) + f(3x + 29) = 0 \quad \forall x \in \mathbb{R}$, then the period of $f(x)$ is
- 7
 - 8
 - 10
 - 54
17. If $f(a - x) = f(a + x)$ and $f(b - x) = f(b + x) \quad \forall x \in \mathbb{R}$ where $a, b (a > b)$ are constants then the period of $f(x)$ is
- $2a$
 - $2b$
 - $3a$
 - b

18. If $f: R \rightarrow R$ satisfies $f(x+y) = f(x) + f(y)$, for all $x, y \in R$ and $f(1) = 7$, then $\sum_{r=1}^n f(r)$ is equal to

- 1) $\frac{7n}{2}$ 2) $\frac{7(n+1)}{2}$ 3) $7n(n+1)$ 4) $\frac{7n(n+1)}{2}$

19. If $f(x) + 2f\left(\frac{1}{x}\right) = 3x, x \neq 0$, and $S = \{x \in R : f(x) = f(-x)\}$; then

- 1) contains exactly one element 2) contains exactly two element
3) contains more than two element 4) is an empty set

LEVEL-II (ADVANCED)

Single answer type questions

- Let $f: (-1, 1) \rightarrow B$ be a function defined by $f(x) = \tan^{-1} \frac{2x}{1-x^2}$, then f is both one-one and onto, when B is in the interval
a) $\left(0, \frac{\pi}{2}\right)$ b) $\left[0, \frac{\pi}{2}\right)$ c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ d) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- Let $A = \left\{x : 0 \leq x < \frac{\pi}{2}\right\}$ and $f: R \rightarrow A$ is an onto function given by $f(x) = \tan^{-1}(x^2 + x + \lambda)$ where
a) $\lambda > 0$ b) $\lambda \leq \frac{1}{4}$ c) $\lambda \geq \frac{1}{4}$ d) $\lambda \geq \frac{1}{8}$
- Let A be a non-empty set of real numbers and $f: A \rightarrow A$ be such that $f(f(x)) = x$ for all x in A . Then f is
a) a bijection b) one- one but not onto
c) onto but not one-one d) neither one-one nor onto
- If the functions $f(x)$ and $g(x)$ are defined on $R \rightarrow R$ such that $f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}$ and $g(x) = \begin{cases} x, & x \in \text{rational} \\ 0, & x \in \text{irrational} \end{cases}$ then $(f-g)(x)$ is
a) one-one and onto b) neither one-one nor onto
c) one-one but not onto d) onto but not one-one
- Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be two one-one and onto functions such that they are the mirror images of each other about the line $y = a$. If $h(x) = f(x) + g(x)$ then $h(x)$ is
a) one-one onto b) one-one into c) many-one into d) many-one onto
- $f: R \rightarrow R, f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ is
a) one-one & onto b) many-one & onto
c) one-one & into d) many-one & into
- The period of the function $f(x) = [6x + 7] + \cos \pi x - 6x$, where $[.]$ denotes the greatest integer function, is
a) 3 b) 2π c) 2 d) 1

8. The period of $f(x) = [x] + [2x] + [3x] + [4x] + \dots + [nx] - \frac{n(n+1)}{2}x$, where $n \in \mathbb{N}$, is (where $[.]$ represents greatest integer function)
- a) n b) 1 c) $\frac{1}{n}$ d) 2
9. Let $f(x)$ be a real valued function with domain $\mathbb{R}$ such that $f(x+p) = 1 + \left[2 - 3f(x) + 3(f(x))^2 - (f(x))^3 \right]^{1/3}$ holds good $\forall x \in \mathbb{R}$ and for some +ve constant 'p' then the period of $f(x)$ is
- a) $\frac{p}{2}$ b) p c) $2p$ d) $\frac{p}{3}$
10. Let ' f ' be a real valued function defined for all $x \in \mathbb{R}$ such that for some fixed $a > 0$, $f(x+a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$ for all ' x ' then the period of $f(x)$ is
- a) $\frac{a}{4}$ b) $\frac{a}{3}$ c) $2a$ d) $3a$
11. Let $f(x, y)$ be a periodic function, satisfying the condition $f(x, y) = f(2x+2y, 2y-2x) \forall x, y \in \mathbb{R}$ and let $g(x)$ be a function defined as $g(x) = f(2^x, 0)$ Prove that $g(x)$ is a periodic function and find its period
- a) 12 b) 10 c) 8 d) 6
12. f is a function defined as $\sum_{k=1}^n f(a+k) = 16(2^n - 1)$ and $f(x+y) = f(x) \cdot f(y)$ and $f(1) = 2$ then integral value of a
- a) 3 b) 0 c) 2 d) 1
13. Let f and g be real-valued functions such that $f(x+y) + f(x-y) = 2f(x) \cdot g(y) \forall x, y \in \mathbb{R}$.
- Statement 1 :** If $f(x)$ is not identically zero and $|f(x)| \leq 1 \forall x \in \mathbb{R}$, then $|g(y)| \leq 1 \forall y \in \mathbb{R}$
- Statement 2 :** For any two real numbers x and y , $|x+y| \leq |x| + |y|$
- a) Statement 1 is true, statement 2 is true, Statement 2 is correct explanation for statement 1
 b) Statement 1 is true, statement 2 is true, statement 2 is not correct explanation for statement 1
 c) Statement 1 is false, but statement 2 is true
 d) Statement 1 is true, but statement 2 is false
14. The function ' f ' satisfies the functional equation $3f(x) + 2f\left(\frac{x+59}{x-1}\right) = 10x + 30$ for all real $x \neq 1$, then the value of $f(7)$ is
- a) 8 b) 4 c) -8 d) 11
15. A function ' f ' well defined $\forall x, y \in \mathbb{R}$ is such that $f(1) = 2$, $f(2) = 8$ and $f(x+y) - kxy = f(x) + 2y^2$, where ' k ' is some constant then $f(x)$ is
- a) x^2 b) $3x^2$ c) $2x^2$ d) $4x^2$

16. A function $F(x)$ satisfies the functional equation $x^2 F(x) + F(1-x) = 2x - x^4$ for all real x . $F(x)$ must be
 a) x^2 b) $1-x^2$ c) $1+x^2$ d) x^2+x+1
17. Consider a real valued function $f(x)$ satisfying $2f(xy) = (f(x))^y + (f(y))^x \quad \forall x, y \in R$ and $f(1)=a$ where $a \neq 1$ then $(a-1) \sum_{i=1}^n f(i) =$
 a) $a^{n+1} + a$ b) a^{n+1} c) $a^{n+1} - a$ d) $a^{n+2} - a$
18. If ' p ' and ' q ' are +ve integers, f is a function defined for +ve numbers and attains only +ve values such that $f(x, f(y)) = x^p y^q$ then $p^2 =$
 a) $2q$ b) q c) $3q$ d) $4q$

More than one correct answer type questions

19. If the following functions are defined from $[-1, 1]$ to $[-1, 1]$, select those which are not objective.
 a) $\sin(\sin^{-1} x)$ b) $\frac{2}{\pi} \sin^{-1}(\sin x)$
 c) $(\operatorname{sgn}(x)) \ln(e^x)$ d) $x^3 (\operatorname{sgn}(x))$
20. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow R$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then
 a) $f(x)$ is an odd function b) $f(x)$ is an one-one function
 c) $f(x)$ is an onto function d) $f(x)$ is an even function
21. Let $f(x) + f(y) = f\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$ [$f(x)$ is not identically zero]. Then
 a) $f(4x^3 - 3x) + 3f(x) = 0$ b) $f(4x^3 - 3x) = 3f(x)$
 c) $f(2x\sqrt{1-x^2}) + 2f(x) = 0$ d) $f(2x\sqrt{1-x^2}) = 2f(x)$
22. $f: R \rightarrow [-1, \infty)$ and $f(x) = \ln([|\sin 2x| + |\cos 2x|])$ (where $[.]$ is the greatest integr function) Then,
 a) $f(x)$ has range Z
 b) $f(x)$ is periodic with fundamental period $\frac{\pi}{4}$
 c) $f(x)$ is invertible in $\left[0, \frac{\pi}{4}\right]$
 d) $f(x)$ is into function
23. Let $f(x)$ be a polynomial with positive degree satisfying the relation
 $f(x)f(y) = f(x) + f(y) + f(xy) - 2$ and $f(4) = 65$, then
 a) $f(2) = 9$ b) $f(3\sqrt{3}) = 27$ c) $f(5) = 126$
 d) The roots of the equation $f(x) = 2x^2$ are all real

24. Let $f: R \rightarrow R$ be a function defined $f(x+1) = \frac{f(x)-5}{f(x)-3} \forall x \in R$. Then which of the following statement(s) is/are true ?
- a) $f(2008) = f(2004)$ b) $f(2006) = f(2010)$
 c) $f(2006) = f(2002)$ d) $f(2006) = f(2018)$
25. The linear function which is defined from $[-1, 1]$ onto $[0, 2]$ is
- a) $f(x) = 1+x$ b) $f(x) = 2-x$ c) $f(x) = 1-x$ d) $f(x) = 3-x$
26. Let $f(x) = [x]^2 + [x+1] - 3$ where $[x] \leq x$. Then
- a) $f(x)$ is many-one and into fun,
 b) $f(x) = 0$ for infinite number of values of x
 c) $f(x) = 0$ for only two real values
 d) $f(x)$ is a one-one function
27. $f(x) = \sin \alpha + \cos \alpha - 1$, where $\alpha = \sin^{-1} \sqrt{\{x\}}, \{.\}$ is the fractional part of x , then $f(x)$ is
- a) an even function b) an odd functions
 c) a periodic function d) zero, $x \in Z$
28. If $f(x)$ is an even function and $g(x)$ is an odd function and satisfies the relation $x^2 f(x) - 2f\left(\frac{1}{x}\right) = g(x)$ then
- a) $f(x) = 0, \forall x \in R$ b) $f(2009) = 0$
 c) f is constant function d) $g(x) = 0, \forall x \in R$

Linked comprehension type questions**Passage - I :**

If $f(x) = \tan x - x$ then

29. The least +ve value of 'x' for which $f(x)=0$ lies in the quadrant
- a) Q_1 b) Q_2 c) Q_3 d) Q_4
30. The set of all the values of $x \in (0, 2\pi)$ for which $f(x) > 0$ is
- a) $(0, \pi)$ b) $(\pi, 2\pi)$ c) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ d) None

Passage - II :

Let $f: R \rightarrow R$ be a continuous function such that $f(x) - 2f\left(\frac{x}{2}\right) + f\left(\frac{x}{4}\right) = x^2$.
 Now answer the following

31. $f(3) =$
- a) $f(0)$ b) $4 + f(0)$ c) $9 + f(0)$ d) $16 + f(0)$
32. The equation $f(x) - x - f(0) = 0$ have exactly
- a) no solution b) one solution c) two solution d) infinite solutions

33. $f(0) =$

a) 0

b) 1

c) $f(0)$

d) $-f(0)$

Matrix matching type questions

34. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be functions such that $f(g(x))$ is a one-one function

Column - I

A) Then $g(x)$

B) Then $f(x)$

C) If $g(x)$ is onto, then $f(x)$

D) If $g(x)$ is into, then $f(x)$

Column - II

p) must be one - one

q) may not be one-one

r) may be many - one

s) must be many one

35. The function $f(x)$ is defined on the interval $[0, 1]$. Then match the following.

Column - I

A) $f: \mathbb{R} \rightarrow \left[\frac{3\pi}{4}, \pi\right)$ and $f(x) = \cot^{-1}(2x - x^2 - 2)$ then $f(x)$ is

B) $f: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = e^{ax} \sin(bx)$ where $a, b \in \mathbb{R}^+$ then $f(x)$ is

C) $f: \mathbb{R}^+ \rightarrow [4, \infty)$ and $f(x) = 4 + 3x^2$ then $f(x)$ is

D) $f: X \rightarrow X$ and $f(f(x)) = x \forall x \in X$ then $f(x)$ is

Column - II

p) one-one

q) into

r) many-one

s) onto

36. Column - I gives functions and column 2 the nature of the functions

Column - I

A) $f: [0, \infty) \rightarrow [0, \infty), f(x) = \frac{x}{1+x}$

B) $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}, f(x) = x - \frac{1}{x}$

C) $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}, f(x) = x + \frac{1}{x}$

D) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + \sin x$

Column - II

p) one - one onto

q) one - one but not onto

r) onto but not one -one

s) neither one - one nor onto

Integer answer type questions

37. If the periodic function $f(x)$ satisfies the equation $f(x+1) + f(x-1) = \sqrt{3}f(x) \quad \forall x \in \mathbb{R}$ and the period of $f(x)$ is 4λ then $\lambda =$

38. If ' f ' is a polynomial such that $f\left(\frac{1-x}{1+x}\right)f\left(\frac{1+x}{1-x}\right) = f\left(\frac{1-x}{1+x}\right) + f\left(\frac{1+x}{1-x}\right)$

(where $x \neq 0, \pm 1$) and $f(3)=28$ then the value of $\frac{1}{605} \left(\sum_{n=1}^{10} (f(n)-1) \right)$ is

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

- 1) 1 2) 2 3) 1 4) 2 5) 3 6) 2 7) 4 8) 1
 9) 2 10) 3 11) 4 12) 3 13) 2 14) 3 15) 4 16) 3
 17) 2 18) 4 19) 4

LEVEL - II

- 1) a 2) d 3) b 4) a 5) d 6) d 7) c 8) d
 9) c 10) b 11) c 12) a 13) d 14) d 15) c 16) b
 17) c 18) c 19) b 20) acd 21) abcd 22) abcd 23) ab 24) ab
 25) abc 26) b 27) ad 28) b 29) a 30) A-iii; B-iv; C-ii; D-v
 31) A-r; B-r; C-p; D-q 32) A-q; B-r; C-s; D-p 33) A-q; B-s; C-r; D-p
 34) 1

EXERCISE-II

LEVEL - I

- 1) 2 2) 4 3) 4 4) 4 5) 2 6) 3 7) 3 8) 4
 9) 1 10) 2 11) 2 12) 2 13) 2 14) 4 15) 1 16) 3
 17) 1 18) 2 19) 2 20) 1 21) 2 22) 3 23) 2

LEVEL - II

- 1) a 2) a 3) c 4) b 5) c 6) c 7) a 8) b
 9) c 10) b 11) d 12) c 13) b 14) a 15) b 16) a
 17) c 18) c 19) c 20) d 21) b 22) b 23) d 24) bc
 25) ab 26) abc 27) ac 28) ad 29) ab 30) c 31) c 32) b
 33) c 34) d 35) c 36) d 37) d 38) a 39) d 40) c
 41) 8 42) 6

EXERCISE-III

LEVEL - I

- 1) 4 2) 1 3) 1 4) 4 5) 3 6) 3 7) 4 8) 4
 9) 1 10) 3 11) 2 12) 4 13) 4 14) 1 15) 3 16) 4
 17) 2 18) 4 19) 2

LEVEL - II

- 1) d 2) c 3) a 4) a 5) c 6) d 7) c 8) b
 9) c 10) c 11) a 12) a 13) a 14) b 15) c 16) b
 17) c 18) b 19) bcd 20) abc 21) ad 22) bd 23) acd 24) abcd
 25) ac 26) ab 27) acd 28) abc 29) c 30) d 31) d 32) c
 33) a 34) A-p; B-q,r; C-p; d-q,r 35) A-q,r; B-r,q; C-p,q; D-p,s
 36) A-q; B-r; C-s; D-p 37) 3 38) 5